



COURSE DESCRIPTION FORM: CL-1002: Programming Fundamentals - Lab

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad Campus

PROGRAM TO BE EVALUATED BS-CS: Fall-2023

Course Description

Lab Course Code	CL-1002		
Lab Course Title	Programming Fundamentals – Lab		
Credit Hours	1		
Lab Course Instructors	Ms. Amna Hassan, Ms. Sher Bano, Ms. Ghulam Fatima		
Grading Policy	Absolute Grading		
Policy about missed assessment items in the course	Retake of missed assessment items (other than sessional/ final exam) will not be held. Student who misses an assessment item (other than sessional / final exam) is awarded zero marks in that assessment item i.e. late submission will not be accepted. For missed sessional/ final exam, exam retake/ pretake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decides the exam retake/ pretake cases.		
Course Plagiarism Policy	Plagiarism in project or midterm/final exam will result in F grade in the course. Plagiarism in any assessment will result in zero marks in that whole assessment category.		
Prerequisites by Course(s) or Topics			
Assessment Instruments with Weights (Sessional exam, final exam, lab tasks, lab project, etc.)	Assessment with the weight.		
	Assessment Item	Number	Weight (%)
	Lab Tasks	>=14	30
	Sessional	1	20
	Project	1	10
	Final Exam	1	40
Lab Course Coordinator	Ms. Amna Hassan		
URL (if any)			

Course Catalog Description	The course aims to equip students with the basic computing concepts and to provide them the ability to analyze the given requirements for solving problems in different domains while implementing the solutions on a computer system. It emphasizes on developing an algorithm and applying the basic programming constructs like control structures, arrays, functions, pointers, dynamic memory allocation, etc. for its development. The students will learn the syntax of the C++ programming language for the implementation.																													
Laboratory Manual	Uploaded on LMS																													
Course Goals	<table border="1"> <tr> <th align="left" colspan="3">A. Course Learning Outcomes (CLOs)</th> </tr> <tr> <td colspan="3"><i>After completion of the course, the students shall be able to:</i></td> </tr> <tr> <td colspan="3"> - Write and Execute Basic C++ Programs following coding conventions using Fundamental Programming Concepts - Utilize Control Structures such as if statements, loops (while, for), and switch-case, to control program flow and make decisions. - Create and call user-defined functions while Understanding function prototypes and the scope of variables. - Manipulate 1D, 2D and 3D Arrays along with understanding Pointers and Memory Management: - Analyze and solve programming problems by breaking them down into smaller, manageable tasks. </td> </tr> <tr> <th align="left" colspan="3">B. Program Learning Outcomes</th> </tr> <tr> <td colspan="3">For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.</td> </tr> <tr> <td>1. Academic Education</td><td>Completion of an accredited program of study designed to prepare graduates as computing professionals</td><td></td> </tr> <tr> <td>2. Knowledge for Solving Computing Problems</td><td>Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems</td><td align="center">✓</td> </tr> <tr> <td>3. Problem Analysis:</td><td>Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.</td><td align="center">✓</td> </tr> <tr> <td>4 Design/Develop Solutions:</td><td>Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.</td><td align="center">✓</td> </tr> </table>			A. Course Learning Outcomes (CLOs)			<i>After completion of the course, the students shall be able to:</i>			- Write and Execute Basic C++ Programs following coding conventions using Fundamental Programming Concepts - Utilize Control Structures such as if statements, loops (while, for), and switch-case, to control program flow and make decisions. - Create and call user-defined functions while Understanding function prototypes and the scope of variables. - Manipulate 1D, 2D and 3D Arrays along with understanding Pointers and Memory Management: - Analyze and solve programming problems by breaking them down into smaller, manageable tasks.			B. Program Learning Outcomes			For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.			1. Academic Education	Completion of an accredited program of study designed to prepare graduates as computing professionals		2. Knowledge for Solving Computing Problems	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems	✓	3. Problem Analysis:	Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.	✓	4 Design/Develop Solutions:	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	✓
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	5. Modern Tool Usage:	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.																																																																															
	6. Individual and Team work:	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.																																																																															
	7. Communication	Communicate effectively on complex computing activities with the computing community and with society at large																																																																															
	8. Computing Professionalism and Society	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.																																																																															
	9. Ethics:	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.																																																																															
	10. Lifelong Learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.																																																																															
	C. Mapping of CLOs on PLOs (CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)																																																																																
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1. Topics to be covered:																																																																																	
Topics covered in the course (assume 15-week instruction and 3 contact hours of lab per week)	List of Topics		No. of Weeks	Contact Hours	CLO(s)																																																																												
	Ubuntu installation, Shell commands		1	3																																																																													
	Pseudo Code And Scratch		1	3																																																																													

	Basic program writing in C++ and stream insertion/extraction operators and dataTypes	1	3	
	Operators (arithmetic, bitwise)	1	3	
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	Sessional-I			
	Conditional structures-II (ternary, switch-case)	1	3	
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	Repetitions-I (while loop, for loop)	1	3	
	Repetitions-II (do-while loop, nested while loop, nested for loop)	1	3	
	Functions (definition, calling, forward declaration)	1	3	
	Arrays-I (1D arrays)	1	3	
	Sessional-II			
	Arrays-II (char and multi-dimensional arrays)	1	3	
	Functions (parameter passing by value/reference)	1	3	
	Recursion	1	3	
	Introduction to pointers and dynamic memory allocation (for 1D)	1	3	
	Total	15	45	
Practical/ Programming Work/ Tools	Ubuntu, Ubuntu shell, Text Editor, g++			
Lab Time Spent (in percentage)	Theory	Problem Analysis & Design	Implementati on	Social and Ethical Issues
	5	15	70	10